

Development in quantum technology will lead to fantastic leaps in fields like materials engineering, drug discovery, understanding of chemical processes (battery development), climate modelling, financial analysis, and cryptography (security).

Metaverse

The metaverse is a hot buzzword today, and everyone is talking about this virtual, parallel universe, and what it is capable of achieving—a thorough merging of the physical and virtual worlds with people seamlessly travelling through these two realms, collaborating with people across the globe, conducting meetings, trainings, research, events, and sales through their *avatars*, and so on and so forth. This kind of full-fledged adoption seems implausible right now. But the fact is that precursor technologies to the metaverse are developing very, very fast. Super-powerful graphics processing units (GPUs), artificial intelligence (AI), cloud platforms, virtual reality (VR) and augmented reality (AR) glasses, high-bandwidth low-latency 5G networks, blockchain technology and decentralised finance (DeFi) are all developing in tandem, accelerating the evolution of this new virtual world.

Interestingly, the metaverse is not just a gamer's paradise. We also see myriad companies, ranging from automakers and banks to hospitals and auctioneers developing their virtual spaces, as a means of engaging with the new generation of tech-savvy customers, to showcase their products and services, and to even sell virtual merchandise. Brands (including celebrities) are selling collectibles in the form of non-fungible tokens (NFTs), which enthusiasts are buying eagerly, hoping to rake in money as their value increases in the future.

On the other hand, several industries including manufacturing,



Inside one of IBM's quantum computers (Courtesy: IBM Research)

pharmaceuticals, logistics, etc are beginning to use metaverse technologies to improve training, collaboration, supply chain and assembly line monitoring and optimisation, and so on. This application, known as the industrial metaverse, is expected to gain a lot of traction this year.

Digital twins

Digital twin technology is emerging as a key aspect of Industry 4.0. It is one of the components of the industrial metaverse, which is 'here and now.' With several cloud platforms available for the same, from companies like Microsoft and Nvidia, it is becoming possible for firms small and large to adopt this interesting technology.

A digital twin is basically a virtual representation of a physical entity—could be a component, an asset, a system or a process. It generally covers the entire lifecycle of the entity, and represents its current state, which is the main differentiator here. The digital twin could be just software, or also include a 2D or 3D visual representation of the entity. It is empowered with real-time information from the physical

entity. So, essentially, the three aspects of this technology are: a physical entity equipped with sensors, a virtual representation of the same, and real-time data exchange between the two.

With advances in edge computing, companies are now beginning to interlink digital twins to get an enterprise-wide picture, which will empower them to optimise their systems and processes and make better decisions.

Cobots and autonomous delivery systems

Cobots or collaborative robots are basically robots that are safe to interact with people, in close proximity. They are not bulky, super powerful robots that have to be confined to cages in the shop floor. They are sweet, friendly, AI-powered robots that can work along with human workers, to increase their efficiency. Cobots are becoming increasingly popular across industries to deliver food in restaurants, to assist doctors and caregivers at hospitals (digital nurses), to move goods on a factory floor, to relieve human workers of repetitive and unergonomic tasks,